

Name: Asmaa Ahmed Maryah
ID: 811634546

Intelligence Information System
Dr. Mohamed Okasha

Q1: Define Artificial Intelligence and Swarm Intelligence.

1 Machine learning

- **Machine learning (ML)** is a subdiscipline of **AI** and constitutes a group of techniques used to solve data analysis by finding patterns among the data.
- These patterns provide the opportunity for understanding complex health situations (e.g., identify disease risk factors) or predicting future health outcomes (e.g., predict disease prognosis).
- **ML** draws competences from various areas of science including data science, statistics, and optimization.
- This tool is different from traditional programming in that it uses learning algorithms as opposed to traditional algorithms and it utilizes mathematical models driven by probability and statistics and invokes predictions using available data to extrapolate unavailable data
- Learning algorithms can learn from data and by feeding this (input) together with the desired results we can obtain the learning algorithm.
- **ML** can be classified into three learning types:
 1. **supervised learning**
 2. **unsupervised learning**
 3. **semisupervised learning**

2. Swarm Intelligence (SI)

Swarm Intelligence is a subfield of AI that focuses on collective, distributed systems (whether natural or artificial).

- **It is based on the decentralized, self-organized collective behavior of individuals in groups.**
- **The system typically consists of a population of simple agents interacting locally with one another and with their environment.**
- **Although there is no centralized control structure dictating how each individual should behave, local interactions between such agents lead to the emergence of "intelligent" and complex global behavior (such as the movement of bird flocks or ant colonies).**

Q2: Give three examples of drawing inspiration from nature for swarm intelligence techniques.

inspires swarm intelligence techniques:

- **Ant Colony Optimization (ACO):** Inspired by the foraging behavior of ant colonies, specifically how ants find the shortest path between their nest and a food source by depositing and following chemical trails called pheromones.
- **Particle Swarm Optimization (PSO):** Modeled after the social behavior and movement patterns seen in bird flocking or fish schooling, where individuals adjust their position based on their own experience and the success of their neighbors to reach a goal.
- **Artificial Bee Colony (ABC):** Derived from the sophisticated honey-gathering behaviors of bee swarms, focusing on how different types of bees (employed, onlookers, and scouts) communicate and coordinate to find the best nectar sources.

Q3: What is meant by optimal distance for Flocks of birds and Schools of fish?

The Concept of Balance: Each individual in a flock or school follows simple local rules to stay close enough to the group for protection and information sharing, but far enough away to avoid collisions.

The "Zone of Repulsion": If a neighbor gets too close (closer than the optimal distance), the individual moves away to maintain personal space and safety.

The "Zone of Attraction": If a neighbor is too far away, the individual moves toward the group to maintain the benefits of the swarm, such as reduced wind resistance or predator defense.

Optimal Stability: The optimal distance is the "sweet spot" where these two forces—attraction and repulsion—cancel each other out, resulting in a stable, organized formation that allows the entire group to move as a single, cohesive unit.

